

Synthesis and Antibiofouling Properties of Novel Crosslinkable Terpolymers Containing Semifluoroalkyl Substituted Aromatic Side Chains

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Synthesis, contact angle analysis, surface properties and biofouling characteristics of novel crosslinkable terpolymers with semifluoroalkyl substituted aromatic side chains have been described. These polymers are targeted for use as coatings to prevent marine biofouling. The marine antifouling properties of these materials were evaluated by laboratory assays employing the fouling diatom *Nitzschia* and ubiquitous *Staphylococcus aureus*. Results indicated that the experimental coatings exhibited better antibiofouling performance than that of a standard Poly(dimethyl siloxane) (PDMS) coating. POLYM. ENG. SCI., 50:944–951, 2010. © 2009 Society of Plastics Engineers

INTRODUCTION

Biofouling on underwater engineered structures, especially on ship hulls, results in increased operational and maintenance costs [1]. Marine biofouling is mainly formed by the accumulating colonization of fouling organisms such as marine bacterium, diatoms. Presently, the most effective approach for prevention of marine biofouling is the use of antifouling paints and coatings which function through the release of toxic ingredients such as cuprous oxide and organotin compounds. But the environmental impacts of long term leaching of toxins in the immediate vicinity of ship become a serious concern [2]. Hence, novel environmentally benign solutions to control biofouling are being sought with urgency.

Apart from a biomimetic approach to protect surfaces from fouling, such as phosphorylcholine (PC)-based polymers having been investigated to establish their potential for resisting biofouling [3, 4], another environmentally benign approach is to degrade an organism's ability to generate a strong interfacial bond with the surface. In this regard, the potency of fluoropolymers and silicon elastomers, which have been considered as the most promising candidates, have been explained in terms of their low surface energy and interesting mechanical properties. Poly(dimethyl siloxane)(PDMS) elastomers become a class of commercial available antibiofouling coating materials owing to their low modulus and low surface energy [5], but the efficacy are lower than those of toxic paints, and regular mechanical cleaning (scrubbing) of the coated surface is required which increased operational expense [6]. Furthermore, it is well known that marine biofilms dominated by diatoms are not released easily from PDMS-based fouling-release coatings [7]. In this respect, fluoropolymer, as another class of low surface energy coating materials, showed better antifouling performance comparing with PDMS coatings; however, these materials often undergo rapid molecular reconstruction in water, and thus lose their fouling release characteristics [8]. For creating a long-term stable fluorinated surface, the fluorinated side chain of a polymer should be able to organize into an order structure on the surface and at the same time to keep $-\text{CF}_3$ groups on the coating surface in a high density [9–12]. To achieve this aim, Ober and co-workers demonstrated that is necessary to keep copolymers bearing semifluorinated alkyl substituted aromatic side chains ($-\text{C}_6\text{H}_5\text{CH}_2\text{O}(\text{CH}_2)_m(\text{CF}_2)_n\text{F}$, $m > 1$, $n > 3$)

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[13] or long semifluorinated alkyl side chains composed of eight CF_2 units and more than four CH_2 units [9–12]. Recently another method found to be effective for creating a stable, nonreconstructable coating was to restrict chain mobility of the polymer through chemical crosslinking group [14], such as epoxide functional group. As demonstrated by experimental results, crosslinking density has a significant influence on the contact angle hysteresis, the lower CA hysteresis was observed (about $20\text{--}44^\circ$) after the photolytic acid-catalyzed epoxide ring-opening cure, which indicated the successful formation of stable or less reconstructable surfaces.

In this article, a class of novel terpolymers (Scheme 3) containing semifluoroalkyl substituted aromatic side chains and crosslinkable epoxide groups was designed and synthesized. The contact angle analysis and surface properties of these fluorinated terpolymer coatings were investigated.

EXPERIMENTAL

Materials

Tetrahydrofuran (THF) was freshly distilled under nitrogen over sodium. Toluene was distilled from CaH_2 . Azo-bis-isobutyronitrile (AIBN) was recrystallized from ethanol and stored under vacuum in the dark until used. Other reagents or materials were used as purchased.

Synthesis of 1H,1H,2H,2H,3H,3H-Perfluoro-1-undecanol (2a)

1H,1H,2H,2H,3H,3H-Perfluoro-1-undecanol has been prepared by a slightly modified procedure previously described by our laboratory [15]. Allyl alcohol (7.4 g, 127.5 mmol) was added dropwise to a mixture of perfluorooctyl iodide (49.8 g, 91.5 mmol) and AIBN (0.36 g, 2.25 mmol) at 95°C . The reaction mixture was stirred for 24 h at 95°C . Then the excessive allyl alcohol was removed in vacuo. Purification of the yellow residue by chromatography on silica gel (petroleum ether:ethyl acetate = 10:1) gave compound **1a** (41.7 g, 75% yield) as a white solid. Then to a mixture of **1a** (33.5 g, 55.5 mmol) and AIBN (0.41 g, 2.5 mmol) was added dropwise tributyltin hydride (18.4 g, 63.5 mmol) at 100°C for 4 h under nitrogen atmosphere. The reaction mixture was stirred at 80°C for 24 h. The phase separation phenomenon occurred. The lower liquid phase was collected and purified by flash chromatography on silica gel (petroleum ether:ethyl acetate = 10:3) to afford compound **2a** (19.4 g, 75% yield) as a white solid. IR (KBr, cm^{-1}): 3369, 2963, 2910, 1377, 1244, 1138. ^1H NMR (CDCl_3 , ppm): δ 3.70 (t, J = 6.0 Hz, 2H), 2.14–2.25 (m, 2H), 1.77–1.83 (m, 2H), 1.50 (s, 1H). ^{19}F NMR (CDCl_3 , ppm): δ –80.72 (t, J = 9.9 Hz, 3F), –114.20 to –114.26 (m, 2F),

–121.66 to –121.87 (m, 6F), –122.65 (s, 2F), –123.46 (s, 2F), –126.06 (s, 2F).

Synthesis of 1H,1H,2H,2H,3H,3H-Perfluoro-1-tridecanol (2b)

Compound **2b** was prepared as described for **2a** (77% yield) with one exception: the anhydrous solvent toluene was added to the reaction system in the second step.

2b, IR (KBr, cm^{-1}): 3344, 2921, 2856, 1431, 1210, 1150. ^1H NMR (CDCl_3 , ppm): δ 3.75 (t, J = 6.1 Hz, 2H), 2.16–2.27 (m, 2H), 1.85–1.90 (m, 2H), 1.57 (s, 1H). ^{19}F NMR (CDCl_3 , ppm): δ –80.67 (t, J = 12.0 Hz, 3F), –114.19 to –114.25 (m, 2F), –121.65 to –121.87 (m, 10F), –122.62 (s, 2F), –123.43 (s, 2F), –126.06 (s, 2F).

Synthesis of Monomer (3a)

Into a three-necked 500 mL flask fitted with a mechanical stirrer and dropping funnel was placed dry THF (200 mL) and compound **2a** (19.4 g, 40.6 mmol). NaH (3.3 g; 60% w/w in mineral oil, 82.5 mmol; hexane washed to remove oil) was added to the rapidly stirring solution, and the reaction mixture was stirred at room temperature for 1 h. The flask was immersed in a cold water bath (10°C), and 4-vinylbenzyl chloride (6.70 g, 43.9 mmol) was added dropwise as such a rate to keep the temperature of the reaction below 25°C . Then the mixture was stirred in the dark at room temperature for further 36 h. It was then concentrated in vacuo at 55°C to ca. 10 mL of solution. Water (50 mL) was added, and the solution was extracted three times with 70 mL of diethyl ether. The combined ether extracts were concentrated in vacuo to give a light yellow oil that was purified by flash column chromatography on silica gel (petroleum ether/ethyl acetate = 30:1) to afford **3a** (16.64 g, 69% yield) as white solid. IR (KBr, cm^{-1}): 3090, 3020, 1630, 1512, 1452, 1241, 1149. ^1H NMR (CDCl_3 , ppm): δ 7.39 (d, J = 8.0 Hz, 2H), 7.27 (d, J = 8.0 Hz, 2H), 6.71 (dd, J_1 = 17.6 Hz, J_2 = 10.9 Hz, 1H), 5.76 (d, J = 17.6 Hz, 1H), 5.25 (d, J = 10.9 Hz, 1H), 4.49 (s, 2H), 3.52 (t, J = 6.0 Hz, 2H), 2.16–2.22 (m, 2H), 1.86–1.92 (m, 2H). ^{19}F NMR (CDCl_3 , ppm): δ –80.79 (t, J = 12.0 Hz, 3F), –114.28 to –114.46 (m, 2F), –121.69 to –121.93 (m, 6F), –122.70 (s, 2F), –123.45 (s, 2F), –126.11 (s, 2F). MS (EI): m/z 594 [M^+]. Anal. Calcd for $\text{C}_{20}\text{H}_{15}\text{F}_7\text{O}$: C, 40.42%; H, 2.54%. Found: C, 41.16%; H, 2.64%.

Synthesis of Monomer (3b)

Compound **3b** were prepared as described for **3a** with one exception: The eluant for the chromatography step was petroleum, followed by petroleum ether/ethyl acetate (10:1): 64% yield. IR (KBr, cm^{-1}): 3091, 3022, 1631, 1513, 1452, 1207, 1148. ^1H NMR (CDCl_3 , ppm): δ 7.39 (d, J = 8.0 Hz, 2H), 7.27 (d, J = 8.0 Hz, 1H), 6.72 (dd, J_1 = 17.6 Hz, J_2 = 10.9 Hz, 1H), 5.74 (d, J = 17.6 Hz,

1H), 5.26 (d, $J = 10.9$ Hz, 1H), 4.49 (s, 2H), 3.52 (t, $J = 6.0$ Hz, 2H), 2.14–2.25 (m, 2H), 1.87–1.93 (m, 2H). ^{19}F NMR (CDCl_3 , ppm): δ –80.72 (t, $J = 12.0$ Hz, 3F), –114.23 to –114.30 (m, 2F), –121.63 to –121.76 (m, 10F), –122.61 (s, 2F), –123.39 (s, 2F), –126.0 (s, 2F). MS (EI): m/z 694 [M^+]. Anal. Calcd for $\text{C}_{22}\text{H}_{15}\text{F}_{21}\text{O}$: C, 38.06%; H, 2.18%. Found: C, 38.62%; H, 2.24%.

General Procedure for Copolymerization

Methyl methacrylate-co-glycidyl methacrylate-co-semi-fluoroalkyl substituted styrene terpolymers were prepared from a free radical polymerization under a nitrogen atmosphere. Into a 100 mL three-necked round-bottom flask equipped with a mechanical stirrer, nitrogen gas inlet, and drying tube filled with calcium sulfate was charged methyl methacrylate (MMA) (6 g, 60 mmol), glycidyl methacrylate (GMA) (6 g, 42 mmol), and **3a** (6 g, 10 mmol). This was followed by addition of 40 mL of THF and 1 wt% AIBN relative to monomer **3a**. The solution was purged for 10 min (N_2), heated to 65°C , and stirred for 24 h. The polymer was precipitated in a blender containing a 50/50 (v/v) mixture of water/methanol, filtered, washed successively with excess methanol, and then dried in a vacuum oven overnight to afford polymer **4a** (13.14 g, 73% yield) as a white solid.

Polymer 4a. IR (KBr, cm^{-1}): 2994, 2952, 1735, 1250, 1149. ^1H NMR (CDCl_3 , ppm): δ 7.06, 6.93, 4.33–4.36(br), 3.67, 3.51, 3.42, 3.13, 2.86, 2.76, 2.55, 1.77, 1.59, 0.79–1.3 (broad multiplet). ^{19}F NMR (CDCl_3 , ppm): δ –80.70 (3F), –114.27 (2F), –121.65 to –121.84 (6F), –122.64 (2F), –123.33 (2F), –126.03 (2 F). GPC (THF): $M_n = 18,600$; $M_w = 22,500$; PDI = 1.21.

Polymer 4b. 68% yield. IR (KBr, cm^{-1}): 2994, 2953, 1732, 1244, 1149. ^1H NMR (CDCl_3 , ppm): δ 7.07, 6.91, 4.33–4.37 (br), 3.71, 3.66, 3.43, 3.14, 2.86, 2.76, 2.56, 1.95, 1.78, 1.53, 0.80–1.29 (broad multiplet). ^{19}F NMR (CDCl_3 , ppm): δ –80.69 (3F), –114.28 (2F), –121.65 to –121.80 (10F), –122.61 (2F), –123.31 (2F), –126.02 (2F). GPC (THF): $M_n = 15,300$; $M_w = 18,500$; PDI = 1.21.

Preparation and Characterization of Polymer-Coated Surfaces

The glass slides were cleaned in hot piranha solution (concentrated sulfuric acid: hydrogen peroxide 7:3 v/v), rinsed with distilled water, and spray-dried using nitrogen. The terpolymer (1.0 g) was dissolved in methyl ethyl ketone (4 mL) to afford a clear, homogeneous solution. To this solution 1.5 wt% (relative to polymer) of WPAG 336 (photoacid generator) was added and stirred until fully dissolved. The polymer-coated slides were prepared by spin-coating using the above solution. Following this, the slides were irradiated in the UV from a mercury arc lamp

for 20 min. This is followed by a postexposure bake at 120°C for 6 h until curing is complete.

Contact angles were measured at 25°C using doubly distilled water by an optical contact angle meter (Model OCA 40, Dataphysics). To achieve an equilibrium state, the measurement of contact angles should begin at the 5th minute of contact. Five drops of each liquid were placed at different locations on a horizontal surface, and five readings of contact angle were taken, and the average value of the five readings was taken as the contact angle of the sample surface.

XPS experiments were carried out by using an Axis Ultra multitechnique electron spectrometer (Kratos, UK) with an AlK α X-ray source and a pass energy of 40 eV. The Al anode voltage was 15 keV and the filament current was 20 mA. The pressure in the spectrometer during analysis was typically in the 10^{-9} Torr range. XPS spectra were recorded automatically. Each measurement was taken on a fresh spot of the sample in order to minimize possible beam damage effects.

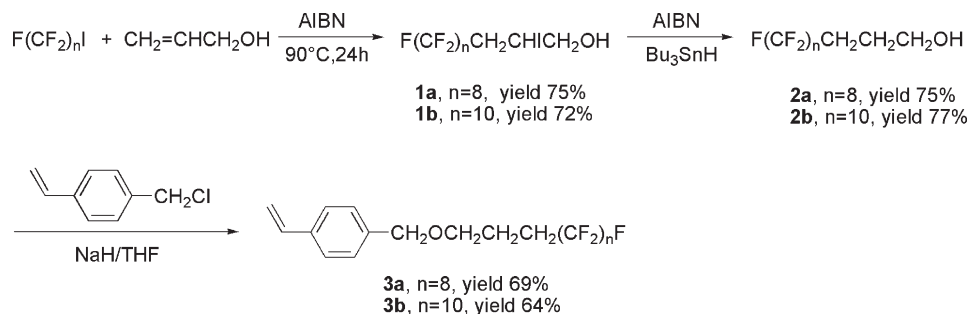
Biofouling Assays

Bacterial Attachment Experiments. In order to assess the ability of the synthesized polymers to resist microorganism attachment, *Staphylococcus aureus* was used for cell adhesion test according to the literature method [16]. The polymer-coated slides were exposed to the bacterial under consideration in an appropriate broth; bacterial attachment was assessed by scanning electron microscope (SEM)(JSM-5600 LV, JEOL).

Settlement and Adhesion Assays with the Diatom Nitzschia. Slides with experiment coating were incubated at about 20°C for 2 days in a 30 L tank of recirculating deionized water. Slides were transferred to seawater for 2 h before the start of the bioassay. *Nitzschia* was obtained from the college of marine life science of Ocean University of China. The diatom *Nitzschia* was cultivated in a climate chamber at $20\text{--}22^\circ\text{C}$ with a light intensity of $49.5\text{--}66\ \mu\text{mol photons m}^{-2}\text{ s}^{-1}$. 10 mL of the cell suspension (1×10^4 cells mL^{-1}) was added to polystyrene culture dishes each containing a glass microscope slide. After 8 days, the slides were very gently washed in seawater to remove cells that had not properly attached. The density of cells attached to the surfaces was counted on each slide using an image analysis system attached to a fluorescence microscope.

Measurements

Elemental analysis was carried out on a Carlo-Erba1106 system. Infrared spectra were obtained on Thermo Electron Corporation Nicolet 380 FT-IR spectrometer. Gas chromatography/mass spectrometry (GC/MS) was recorded on a Finnigan MAT-8430 instrument using EI ionization at 70 eV. ^1H NMR (400 MHz) and



SCHEME 1. Preparation of monomer **3a** and **3b**.

^{19}F NMR (376 MHz) spectra were recorded on a Bruker AM 400 spectrometer system. Chloroform- d was used as the solvents and chemical shifts reported were internally referenced to Me_4Si (0 ppm) and CFCl_3 (0 ppm) for ^1H , ^{19}F NMR, respectively. Relative molecular weights and molecular weight distributions were measured by gel permeation chromatography (GPC) system equipped with a Waters 1515 Isocratic HPLC pump, a Waters 2414 refractive index detector (RI), a Waters 2487 dual-wavelength absorbance detector, and a set of Waters Styragel columns (HR3, HR4, and HR5, 7.8×300 mm). GPC measurements were carried out at 35°C using THF as eluent with a 1.0 mL/min flow rate. The system was calibrated with polystyrene standards. Dynamic thermogravimetric analysis (TGA) was performed on NETZSCH (Germany) TGA 209 F1 system for film or powder samples at a heating rate of $10^\circ\text{C}/\text{min}$ under nitrogen atmosphere from 25°C to 700°C . The water contact angles were measured using an automatic video contact-angle testing apparatus (OCA 40, Dataphysics). A volume ($3 \mu\text{L}$) of the test liquid (distilled water) was applied to slides, and the contact angle was determined from the video camera images of the drop in the course of its formation. The contact angle measurements were performed 5 min after a drop of the test liquid was placed on slides. The structural and surface morphology of the synthesized polymer was recorded by scanning electron microscope (SEM) (JEOL, JSM-5600 LV). The Chemical composition of the treated surface

was characterized by X-ray photoelectron spectroscopy XPS, (XSAM800, Kratos, UK).

RESULTS AND DISCUSSION

Synthesis of Semifluoroalkyl-Substituted Styrene **3**

The semifluoroalkyl-substituted styrene **3a** and **3b** were synthesized in three steps as shown in Scheme 1. Perfluorooctyl iodide (or perfluorodecyl iodide) was converted into **1** through addition reaction with allyl alcohol. Compound **1** was readily oxidized in air, so it was prepared and immediately used without further purification. After reducing the iodide in compound **1a** and **1b** by tributyltin hydride, the resulting semifluorinated alcohol **2a** and **2b** was reacted with the excess 4-vinylbenzyl chloride in the presence of sodium hydride to yield **3a** and **3b**. The composition and structure of compounds **3a** and **3b** were confirmed by MS, FT-IR, ^1H NMR, ^{19}F NMR spectra, and elemental analyses. Figures 1 and 2 depict typical

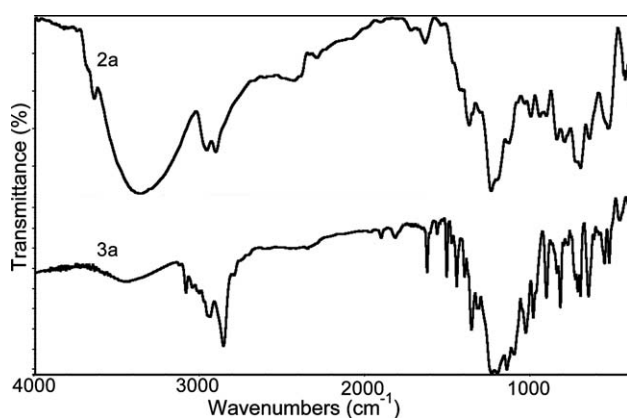


FIG. 1. FTIR spectra of **2a** and **3a**.

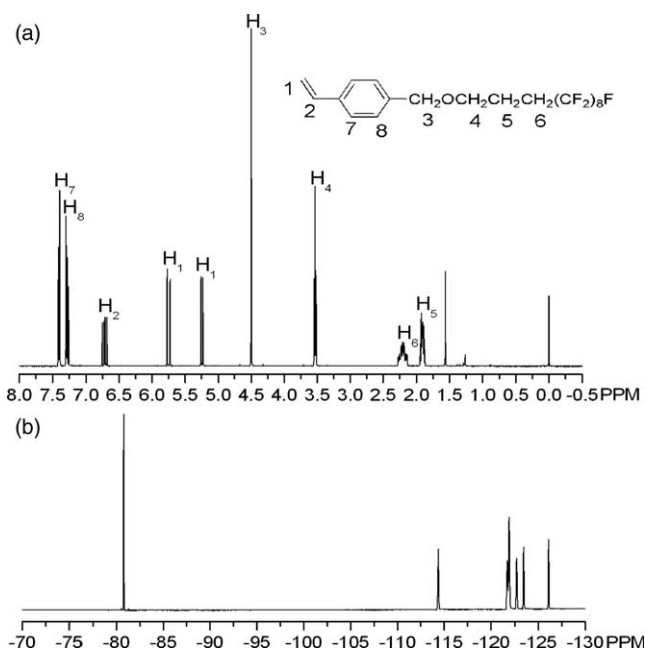
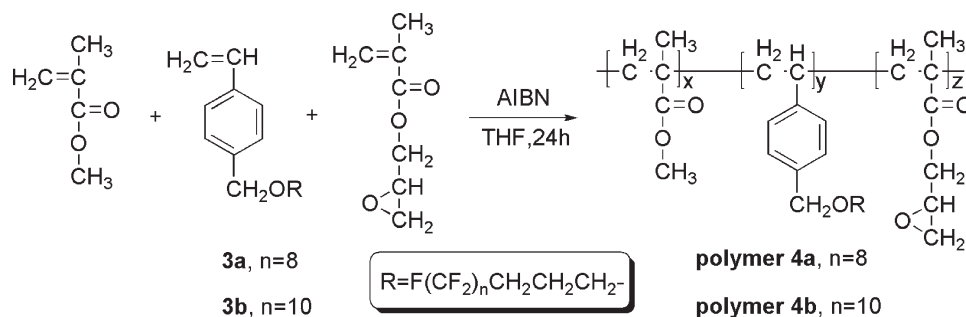


FIG. 2. ^1H NMR (a) and ^{19}F NMR (b) spectra of monomer **3a**.



SCHEME 2. Preparation of the terpolymers **4a** and **4b**.

FTIR, ^1H NMR, ^{19}F NMR spectra of the compound **3a**. In the IR spectra of **2a** and **3a**, the C–F multiple stretch bending absorptions of fluoroalkyl group were observed between 1300 and 1000 cm^{-1} (see Fig. 1). For compound **3a**, the C=C stretching absorption appeared at 1631 cm^{-1} . The disappearing of strong broad absorption band at 3400 cm^{-1} for hydroxyl group on the FTIR spectrum of **3a** indicated that compound **2a** was completely converted. In the ^1H NMR spectrum (Fig. 2a) of **3a**, the aromatic proton signals appeared at 7.39 and 7.27 ppm as two doublet peaks; signals at 6.71 (dd, H_2 , 1H), 5.76 (d, H_1 , 1H) and 5.26 ppm (d, H_1 , 1H) belonged to protons of $\text{H}_2\text{C}=\text{CH}-$; 4.49 (s, H_3 , 2H), 3.52 (t, H_4 , 2H), 2.16–2.22 (m, H_6 , 2H), 1.86–1.92 (m, H_5 , 2H) belonged to $-\text{CH}_2-$ in the semifluorinated alkyl group. In the ^{19}F NMR spectrum (Fig. 2b), fluorine signals in the semifluorinated alkyl group appeared at the range of -80.8 to -126.1 ppm. The successful synthesis of the high-purity monomer **3a** was further confirmed by elemental analysis.

Polymer Synthesis

Terpolymers **4a** and **4b** containing semifluoroalkyl substituted aromatic side chains were synthesized through a free radical copolymerization starting from the described monomers **3a** and **3b** under a nitrogen atmosphere as shown in Scheme 2. Viscous products were formed after reacting for approximate 24 h which indicated low to medium molecular weight polymer was created. The chemical structures of random terpolymers **4a**

and **4b** were characterized by FT-IR, ^1H NMR, ^{19}F NMR, and GPC analyses. In the FT-IR spectra, both polymers showed characteristic carbonyl stretching vibrations at 1730 to 1735 cm^{-1} . The C–F multiple stretching absorptions were also detected in the range of 1250 to 1100 cm^{-1} (see Fig. 3). Figure 4 depicts typical ^1H and ^{19}F NMR spectra of the random polymer **4a**. It can be seen that proton signals at 6.71 (dd, H_2 , 1H), 5.76 (d, H_1 , 1H) and 5.26 ppm (d, H_1 , 1H) for protons of $\text{H}_2\text{C}=\text{CH}-$ group (see Fig. 2) disappeared and new signals appeared at 0.79–1.3 ppm (Fig. 4a), suggesting the successful synthesis of the polymer **4a**. GPC analyses gave M_w values of 22,500 and 18,500 for **4a** and **4b**, respectively. Polymer **4b** containing longer perfluoroalkyl chain has a lower molecular weight than that of **4a**. The values of M_w/M_n are 1.21 for **4a** and **4b** regardless of the length of perfluoroalkyl chains.

Photocure of the Novel Terpolymers **4a** and **4b**

The terpolymer coated films were made from a solution of polymers **4a** or **4b** and 1.5 wt% of WPAG 336 (Scheme 3) in methyl ethylketone. An organic acid which

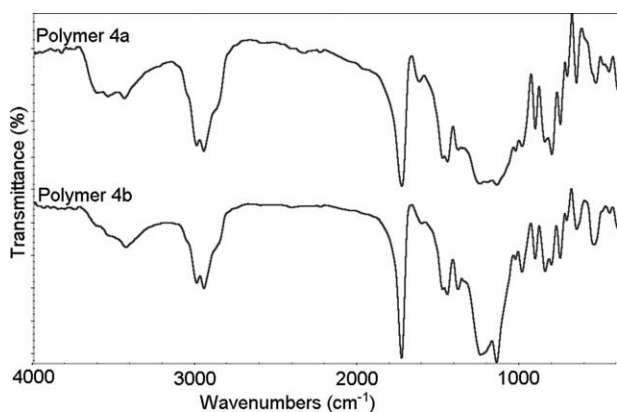


FIG. 3. FTIR spectra of polymers **4a** and **4b**.

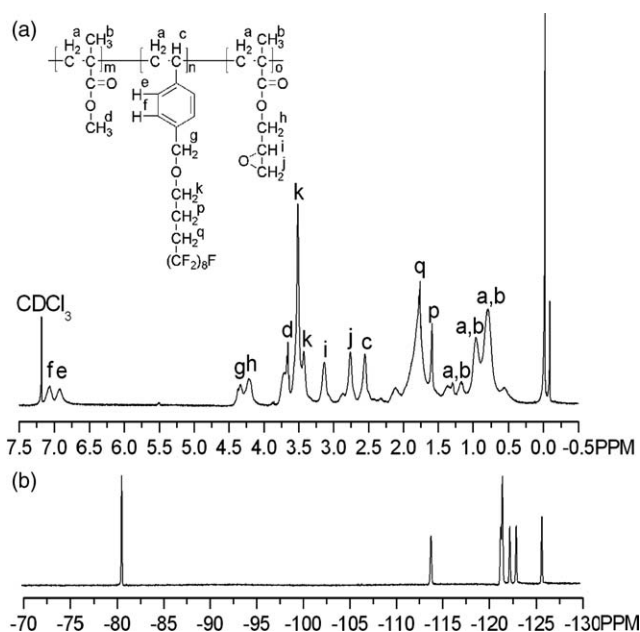
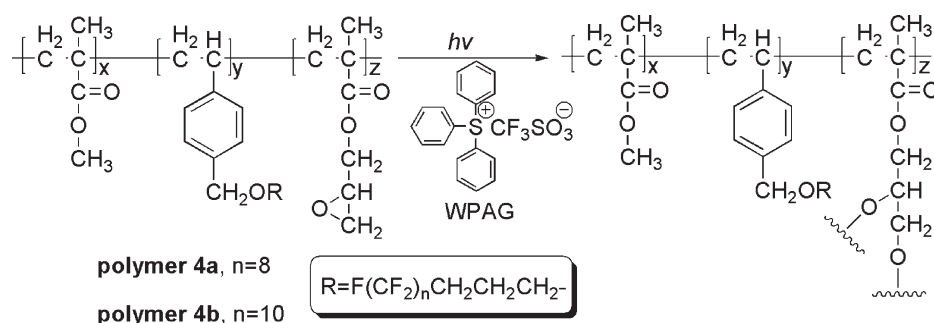


FIG. 4. ^1H NMR (a) and ^{19}F NMR (b) spectra of polymer **4a**.



SCHEME 3. The photolytic acid-catalyzed epoxide ring-opening cure chemistry.

catalyzes the ring-opening of the epoxy pendant groups in the glycidyl methacrylate monomer is generated once irradiated under UV lamp [14].

Water Contact Angle Measurements

Table 1 showed the advancing angles and receding angles of a water drop on the coated surfaces that were used in the biofouling assays. As expected, the advancing angles for polymer **4a** and **4b** coatings were found to be $(116 \pm 1^\circ)$ and $(119 \pm 1^\circ)$, respectively, but a relative low receding angle (about 76°) was observed for **4a** coating. As was demonstrated by Schmidt et al. [17], significant contact angle hysteresis ($\theta_a - \theta_r$) means surface reconstruction. The contact angle hysteresis (40°) for **4a** coatings indicated that slight surface reconstruction was happened on the surfaces of polymer coating and the introduction of aromatic ring in the polymer side chains has not offered much influence in stabilizing the surface structure of polymer coating. The relatively lower contact angle hysteresis (18°) of polymer **4b** coating showed that the influence of surface reconstructing can be partially counterbalanced by the introduction of longer perfluoroalkyl chain.

XPS Analysis

The surface chemical structure of polymer film was analyzed by XPS. Figure 5 shows the typical XPS survey spectra and the detailed spectra of C1s peak of polymer

4a coating surface. C1s, O1s, and F1s peaks were detected at around 285, 535, and 690 eV, respectively (Fig. 5b). Furthermore, apart from the distinct C=O, C—O, C—C, C=C peaks at 288.9, 286.0, and 285 eV, the expected CF₂ and CF₃ peaks were observed at binding energies about 291.7 and 293.9 eV as shown in Fig. 5a. The XPS spectra indicated the successful formation of the polymer coating on the slide surfaces which were used in the biofouling assays. The same results have been seen for polymer **4b**.

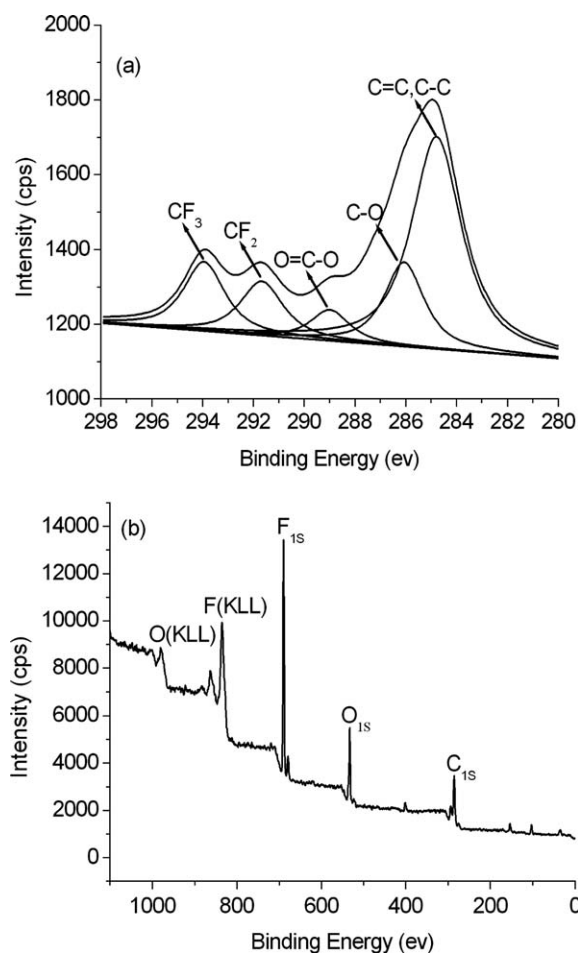


FIG. 5. The XPS carbon 1s spectra (a) and survey spectra (b) of polymer **4a** coating surface.

TABLE 1. The advancing and receding water contact angles of coatings on glass microscope slides, prepared by spin-coating technique.

	Water contact angles (deg)		
	Advancing (θ_a)	Receding (θ_r)	Hysteresis ($\theta_a - \theta_r$)
Film 1 ^a	116 ± 1	76 ± 1	40
Film 2 ^b	119 ± 1	101 ± 1	18

^a Prepared by spin-coating a mixture solution of polymer **4a**, WPAG 336 in methyl ethyl ketone on slides and annealing at 120°C for 6 h.

^b Prepared by spin-coating a mixture solution of polymer **4b**, WPAG 336 in methyl ethyl ketone on slides and annealing at 120°C for 6 h.

TABLE 2. Thermal properties of polymers.

Polymers	T_d (°C) ^a	$T_{5\%}$ (°C) ^b	$T_{10\%}$ (°C) ^b
4a	305	267	288
4b	307	269	294

^a The onset decomposition temperature.^b Temperatures at 5% and 10% weight loss.

Thermal Properties of Polymers

Thermogravimetric analyses of the synthesized polymers were carried out under N₂ atmosphere with a heating rate of 10°C/min. The thermal analytical data of polymer **4a** and **4b** were listed in Table 2. The polymers

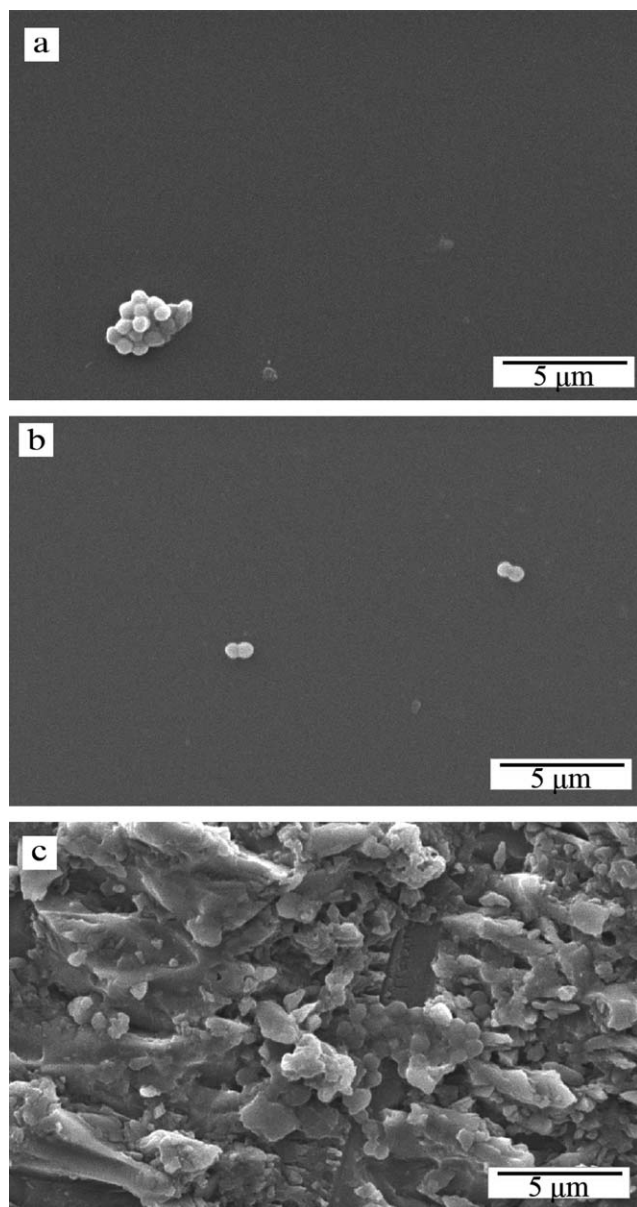


FIG. 6. SEM images of the treated slides (a, b) and the bare glass (c) after exposure to the culture of *Staphylococcus aureus* for 48 h.

exhibited temperatures corresponding to 5% weight loss ranging from 267°C to 269°C and 10% weight loss ranging from 288°C to 294°C. The onset thermal decomposition temperatures (T_d) were higher than 300°C. The results of thermal analysis demonstrated that the resulting crosslinkable terpolymers had good thermal stability.

Biofouling Assays

As we know, bacterial adhesion on surfaces underwater is a prerequisite for further biofouling caused by marine organisms [18]. Therefore, it is necessary to prevent or reduce bacterial adhesion. The comparison of the numbers of bacterial cell colonies attached to the polymer **4a** and **4b** treated slides and the bare glass were shown in Fig. 6. The bacterial colonies and biofouling could be identified on bare substrates obviously by scanning electron microscope (SEM) as shown in Fig. 6c. On the contrary, only slight bacterial colonies were found on the coated slides after exposure to culture of *Staphylococcus aureus* for 48 h (Figs. 6a, 6b). This effect was due to the inability of the bacterial exopolymer to wet or become attached to hydrophobic surfaces, especially for polymer **4b** (advancing contact angle: about 120°). This study suggested that the designed polymer **4** has promising anti-biofouling performance.

Furthermore, the test results of the fouling diatom *Nitzschia* attachment on films of PDMS and crosslinkable polymers **4a** and **4b** were shown in Fig. 7. The number of diatom *Nitzschia* cells on the surface of polymer **4a** and **4b** coatings were 180×10^3 and $150 \times 10^3/\text{cm}^2$, respectively, which were lower than that on commercial available PDMS coating ($380 \times 10^3/\text{cm}^2$). This result indicated that more cells were adhered on the PDMS surface than that on the fluorinated polymer surfaces. In addition, it was found that attachment strength of *Nitzschia* to polymer **4b** is weaker compared with polymer **4a** which

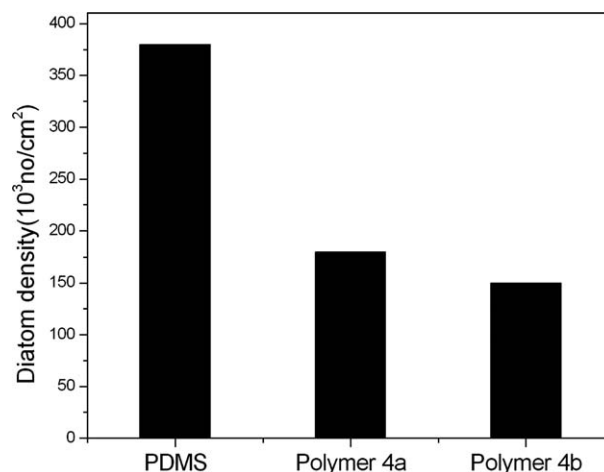


FIG. 7. Settlement of *Nitzschia* on the surfaces of PDMS, Polymers **4a** and **4b** coatings.

is consistent with the result of contact angle hysteresis analyses.

CONCLUSIONS

Novel crosslinkable terpolymers containing semifluor-oalkyl substituted aromatic side chains were prepared through a free radical polymerization. Surfaces of these polymer coatings were well characterized by dynamic thermogravimetric analysis (TGA), XPS, contact angle hysteresis, biofouling assays. Polymers **4a** and **4b** possessed good thermal stability ($T_d > 300^\circ\text{C}$) and relatively low CA hysteresis (about $18\text{--}40^\circ$). The biological data indicated that these materials, especially the polymer **4b**, contain good antibacterial ability and better antifouling property comparing with PDMS elastomers.

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